CMPT 365 Written Assignment 3

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April 5 2024

Section 8.10

Question 4

In a 2D DCT transform of block size (8×8), the total number of coefficients each block is 64 (8×8).

Since we are only using direct current (DC) component to create a thumbnail image, we are essentially taking the first coefficient in each block out of 64.

Each block of our given size only has one DC, so the fraction of original pixels we would be using is 1/64.

Question 5 a)

The largest possible DCT coefficient value occurs when the input image has the maximum possible pixel value for all 8×8 pixels, which is 255 in our given range 0 to 255. In this case, all pixels having value 255 gives a white image.

The largest DCT coefficient would be in the top left DC component, which can find using equation given in 8.17 of textbook. We know DC component means i = 0, j = 0, while C(ξ) = $\frac{\sqrt{2}}{2}$ if $\xi = 0$, C(ξ) = 1 otherwise, where ξ is u, v = 0.

$$= \frac{C(u)C(v)}{4} \sum_{i=0}^{7} \sum_{j=0}^{7} cos \frac{(2i+1)u\pi}{16} cos \frac{(2j+1)v\pi}{16} f(i,j)$$

$$= \frac{\frac{\sqrt{2}}{2}\frac{\sqrt{2}}{2}}{4} \times cos(0) \times cos(0) \times 255$$

$$= \frac{255}{8}$$

Since u, v = 0, we could disregard Cos, $C(u)$, $C(v)$ and simply iterate over i, j. This results in $\frac{255}{8}$. However, we iterated over these values 64 times, so we have $64(\frac{255}{8}) = 2040$, our largest coefficient (DC), while all other values = 0.

Section 9.5

Question 2

Without the luminance component, we cannot completely restore the original color image. Missing the luminance component would result in:

1. A lower apparent brightness of the image, since luminance is responsible for the percieved brightness of a color.
2. Image will look a little more pixelated and blocky, since each block of the image is forced to share same color values.

Question 4 a)

i) Same equation as section 8.10 question 5 a):

$$= \frac{C(0)C(0)}{4} \sum_{i=0}^{7} \sum_{j=0}^{7} cos\frac{0}{16} cos\frac{0}{16} f(i,j)$$

$$= \frac{\frac{\sqrt{2}}{2}\frac{\sqrt{2}}{2}}{4} \sum_{i=0}^{7} \sum_{j=0}^{7} cos\frac{0}{16} cos\frac{0}{16} f(i,j)$$

$$= \frac{1}{8} \times \sum_{i=0}^{7} \sum_{j=0}^{7} f(i,j)$$

$$= \frac{(16 \times 20) + (16 \times 80) + (16 \times 140) + (16 \times 200)}{8}$$

$$= 880$$

Sum of entries for image f(i, j) divided by its size 8 returns DCT value 880.

ii) AC Coefficient would be the largest when u = 1, v = 0 (1, 0). This corresponds to the lowest non zero horizontal frequency (u = 1) and the DC vertical frequency (v = 0). When u = 1, v = 0, cosine terms have the slowest oscillation, therefore resulting in the largest magnitude for the AC coefficient. The sign is negative because repeating pixel value pattern has opposite phase.

Exercise 1

i)

Encoder Pseudo Code

Symbol	Prob.
1	0.8
2	0.02
3	0.18

- Keep track of LOW, HIGH, RANGE
  - Any two are sufficient, e.g., LOW and RANGE.

```
low=0.0, high=1.0;
while (not EOF) {
    n = ReadSymbol();
    RANGE = HIGH - LOW;
    HIGH = LOW + RANGE * CDF(n);
    LOW = LOW + RANGE * CDF(n-1);
}
output LOW;
```

Input	HIGH	LOW	RANGE
Initial	1.0	0.0	1.0
1	0.0 + 1.0 × 0.8 = 0.8	0.0 + 1.0 × 0 = 0.0	0.8
2	0.0 + 0.8 × 0.82 = 0.656	0.0 + 0.8 × 0.8 = 0.64	0.016
3	0.64 + 0.016 × 1.0 = 0.656	0.64 + 0.016 × 0.82 = 0.65312	0.00288
1	0.65312 + 0.00288 × 0.8 = 0.655424	0.65312 + 0.00288 × 0 = 0.65312	0.002304

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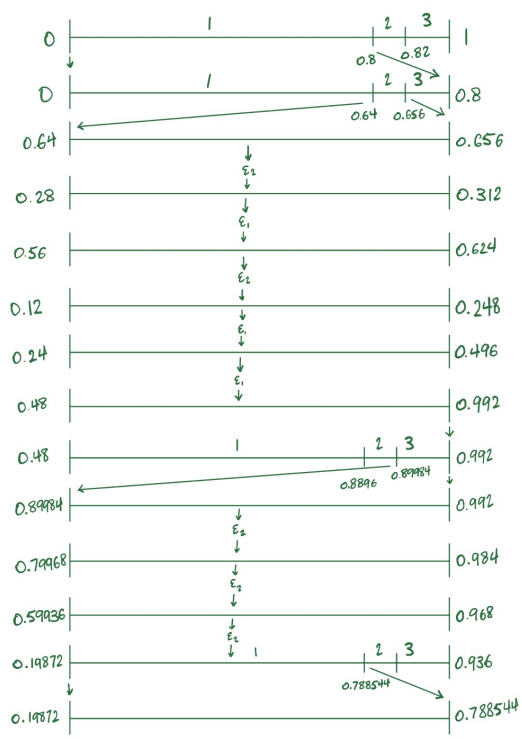
ii) E1/E2 Scaling Operations - E2 E1 E2 E1 E1 E2 E2 E2



Exercise 2

i) (0, 38), (0, 87), (0, 16), (2, 54), (6, 12), (2, 3), (0, 0)
(DC Component (0, 15) is left out according to the textbook)

ii) Corresponding Sequence - 15 38 0 54 0 0 0 0 87 0 0 12 3 0 0 0 16 .....
Runlength Coding - (0, 38), (1, 54), (4, 87), (2, 12), (0, 3), (3, 16), (0, 0)
(DC Component (0, 15) is left out according to the textbook)